

RAGAS vs. cheap local shortcuts: an evaluation on a real agentic-RAG memory store

Author: Alkiviadis (Adi) Lazaridis

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Status: research report — aggregate findings only

Scope: 4 RAGAS metrics vs. 9 deterministic shortcut variants, 99 joined samples, fully local compute

Abstract

LLM-as-a-judge is the default way to evaluate RAG systems, but it is slow, non-deterministic, and opaque. We evaluate whether cheap, deterministic shortcuts built from embeddings, cross-encoders, NLI, and lexical overlap can predict what a local LLM judge would say, on the session store of a real agentic coding tool (opencode) mirrored into a Chroma vector database. We find that **AnswerCorrectness** is strongly predicted by a ROUGE-L + semantic-cosine blend (Spearman $\rho = 0.89$) and **AnswerRelevancy** by nomic query/document cosine ($\rho = 0.71$), meaning the expensive judge is replaceable for ranking. Conversely, **Faithfulness** agreement is weakly negative ($\rho = -0.18$) because the judge saturates at a median of 1.000 while the NLI shortcut reads a median of 0.000, and **ContextPrecision** has no discriminative signal on this corpus (all variants near-constant). We conclude that shortcut-vs-judge agreement is a fast, actionable diagnostic for which RAG metrics are trustworthy on a given corpus — and which judge metrics themselves need an oracle audit.

1. Introduction

Retrieval-augmented generation (RAG) systems are typically evaluated with LLM-as-a-judge frameworks such as RAGAS. These metrics require a capable judge model, are non-deterministic, and take hours on a realistic dataset with local models. For teams that iterate on retrieval pipelines (chunking, embedding models, ranking, context-window policy), a *deterministic* score that predicts the judge would be far cheaper to run in CI.

This study answers a narrower, more useful question than “which metric is best”: **on a given corpus, which RAGAS metric can be replaced by which cheap shortcut, and which metrics are themselves unreliable?**

The subject is not a toy dataset. It is the full session history of an agentic coding assistant — thousands of real conversations across dozens of projects — mirrored into a local vector database. The queries are genuine user requests; the responses are fresh generations conditioned on retrieved chunks; the references are the historical assistant replies. This is exactly the kind of messy, production-shaped data on which benchmark conclusions usually fail.

2. System under evaluation

The vector store under test (opcode_sessions) is derived from a SQLite session database via a two-stage pipeline:

1. **Extract**: session messages are read from the materialized message/part tables and written to a JSONL corpus, chunked to ~1400 chars with overlap.
2. **Embed**: the corpus is embedded with nomic-embed-text (768-d, cosine) on a local GPU via Ollama and stored in Chroma.

Retrieval is top-k cosine search; the generator (gemma4:12b, temperature 0) answers given the top-5 chunks. Section 5 of METHODOLOGY.md documents the architecture in detail.

3. Method

3.1 Dataset

100 query/reference/context triples sampled with seed 42 (≤ 5 per session) from the session store; responses freshly generated at temperature 0 from the top-5 retrieved chunks. 99 samples survive to the joint analysis.

3.2 RAGAS pass

Classic ragas.metrics (Faithfulness, AnswerRelevancy, ContextPrecision, AnswerCorrectness) with a local gemma4:12b judge and nomic-embed-text embeddings. Single worker (Ollama is serial); the full run took 8h26m.

3.3 Shortcut pass

Deterministic, GPU-computable approximations, designed to mirror each metric’s semantics:

RAGAS metric	Shortcut	Implementation
ContextPrecision	graded NDCG	relevance = $\sigma(\text{cross-encoder}(\text{reference}, \text{chunk}))$; order = query-vs-chunk reranker (bge / MiniLM), retrieved & reranked
Faithfulness	sentence NLI	DeBERTa-v3 MNLI entailment ≥ 0.5 per sentence, max over chunks
AnswerRelevancy	TAS-B dot / nomic cosine	asymmetric QA embeddings, prefixed search_query: / search_document:

RAGAS metric	Shortcut	Implementation
AnswerCorrectness	ROUGE-L + cosine	$0.75 \cdot \text{ROUGE-L-F1} + 0.25 \cdot \text{cosine}(\text{response}, \text{reference})$

The threshold-based ContextPrecision shortcut was rejected after it collapsed ($\approx 89\%$ of chunks lexically relevant $\rightarrow \tau \approx 0$); the graded design restores discriminative structure.

3.4 Agreement analysis

Pearson and Spearman correlation over the 99 joined samples per metric; verdict bands at $|\rho| \geq 0.7$ (STRONG), 0.4-0.7 (MODERATE), < 0.4 (WEAK).

4. Results

4.1 AnswerCorrectness — replaceable

Variant	Pearson	Spearman
$0.75 \cdot \text{ROUGE-L} + 0.25 \cdot \text{nomic}$	0.822	0.894
ROUGE-L only	0.788	0.870
nomic only	0.775	0.875

The combined blend tracks the judge almost perfectly in rank. All three variants are STRONG; the blend adds little over either alone (0.89 vs 0.87/0.87), so **any** of them suffices for ranking.

4.2 AnswerRelevancy — strong with the right embedding

Variant	Pearson	Spearman
nomic query/doc cosine	0.777	0.709
TAS-B dot	0.556	0.576

The nomic cosine (same embedding family as the retrieval index) is STRONG; the fine-tuned TAS-B (saturating, unbounded scale) is only MODERATE.

4.3 Faithfulness — judge saturates, shortcut cannot be trusted yet

RAGAS median **1.000**, NLI median **0.000**. Spearman -0.185 (WEAK, negative). The judge calls essentially everything faithful; the NLI shortcut calls most sentences unentailed by the retrieved chunks. With a bimodal judge, the correlation is driven by a few low outliers. The correct response is an oracle audit (human sentence-level entailment labels on 20-30 samples) — not “trust the shortcut”.

4.4 ContextPrecision — no signal on this corpus

Variant	mean NDCG	Spearman
bge (retrieved)	0.990	−0.124
bge (reranked)	0.991	−0.044
MiniLM (retrieved)	0.755	−0.018

Every variant is essentially constant. On this corpus all retrieved chunks are relevant, so the metric cannot discriminate; the only variance comes from the RAGAS judge’s own chunk judgment. The metric is corpus-degenerate here, not wrong in general.

4.5 Aggregate summary

RAGAS metric	Shortcut	ρ (Spearman)	Verdict
AnswerCorrectness	combined	0.894	STRONG
AnswerCorrectness	ROUGE	0.870	STRONG
AnswerCorrectness	nomic	0.875	STRONG
AnswerRelevancy	nomic cosine	0.709	STRONG
AnswerRelevancy	TAS-B	0.576	MODERATE
Faithfulness	NLI	−0.185	WEAK
ContextPrecision	bge/MiniLM	≈ 0	WEAK

5. Discussion

For ranking, cheap wins. Two of four RAGAS metrics are reliably predicted by shortcuts that run in minutes instead of hours and are fully deterministic. The implication for CI: gate AnswerCorrectness and AnswerRelevancy on the shortcuts; run the judge only on periodic calibration runs.

Disagreement is a diagnostic, not an error. Where ρ is near zero the shortcut and judge measure different things. Faithfulness exposes a judge ceiling; ContextPrecision exposes a corpus degeneracy. Both findings are actionable: audit the judge with an oracle, or switch metrics.

Threats to validity. Single judge and generator (both gemma4:12b); a stronger or differently-prefixed judge could shift the correlations. The TAS-B shortcut’s unbounded scale makes only rank statistics meaningful. The corpus has a ceiling on context relevance that may not generalize.

6. Conclusions

1. AnswerCorrectness and AnswerRelevancy can be evaluated cheaply and deterministically on this system.
2. Faithfulness is currently *undecidable* — both judge and shortcut are plausible; an oracle audit is the next step.
3. ContextPrecision has no discriminative power on this corpus; do not gate on it.

4. The shortcut-vs-judge methodology itself is cheap, and worth running before adopting any RAG metric on a new corpus.

7. Reproducibility

src/ contains the full pipeline (reproduce_eval.py, shortcut_metrics.py, compare_eval.py, build_aggregates.py); tests/ covers the pure functions; METHODOLOGY.md documents every parameter. Data and per-sample rows are excluded by policy (data/README.md).

References

- Es et al. (2023), *RAGAS: Automated Evaluation of Retrieval Augmented Generation* (see lit_digest.md for this and other foundational works).